Biostatistical Methods II Final Model Review (Lectures 12-23)

1. Poisson Log-Linear Model (L12)

Model: $Y_i \sim \text{Poisson}(\mu_i)$ independent.

$$\log \mu_i = \mathbf{x}_i^T \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

$$E(Y_i) = \mu_i = \exp(\mathbf{x}_i^T \boldsymbol{\beta}), \;\; \text{Var}(Y_i) = \mu_i$$

Interpretation:

- β_0: log of expected count when all $x = 0$.
- β_j: change in **log expected count** per unit increase in x_j (others fixed).
- $\exp(\beta_j)$: **rate ratio (RR)**. Counts are multiplied by $\exp(\beta_j)$ for a 1-unit increase.
- E.g., $\beta_j = 0.2 \Rightarrow$ counts increase by a factor of $e^{0.2} \approx 1.22$ (22% higher).

2. Poisson Rate Model (with Offset, L12)

Model: Events Y_i over exposure n_i (time/area/persons) at rate λ_i.

$$Y_i \sim \text{Poisson}(\mu_i), \;\; \mu_i = n_i \lambda_i$$

$$\log \mu_i = \underbrace{\log n_i}_{\text{offset}} + \mathbf{x}_i^T \boldsymbol{\beta}$$

Interpretation:

- $\log n_i$ is the **offset** — known, coefficient fixed at 1.
- $\boldsymbol{\beta}$ now describes covariate effects on the **rate** $\lambda_i = \exp(\mathbf{x}_i^T \boldsymbol{\beta})$, NOT on the count.
- $\exp(\beta_j)$: **log rate ratio** — multiplicative change in rate per unit x_j.

3. Quasi-Poisson Model (L13)

Model: Same mean structure as Poisson, but variance allows overdispersion.

$$\log \mu_i = \mathbf{x}_i^T \boldsymbol{\beta}, \;\; E(Y_i) = \mu_i, \;\; \text{Var}(Y_i) = \phi \mu_i, \;\; \phi > 1$$

Interpretation:

- $\boldsymbol{\beta}$: same interpretation as in Poisson ($\hat{\boldsymbol{\beta}}_Q = \hat{\boldsymbol{\beta}}_{MLE}$).
- ϕ: **dispersion parameter**. Estimated by $\hat{\phi} = G/(n-p)$.
- $\text{cov}(\hat{\boldsymbol{\beta}}_Q) = \hat{\phi} \cdot \text{cov}(\hat{\boldsymbol{\beta}}_{MLE})$ — SEs inflated by $\sqrt{\hat{\phi}}$.
- No proper likelihood (quasi-likelihood only).

4. Negative Binomial Regression (L13)

Model: $Y_i \sim NB(\mu_i, \phi)$ with reparameterization

$$P(Y = y) = \frac{\Gamma(y+\phi)}{\Gamma(y+1)\Gamma(\phi)} \frac{\mu^y \phi^\phi}{(\mu+\phi)^{\phi+y}}$$

$$\log \mu_i = \mathbf{x}_i^T \boldsymbol{\beta}, \;\; E(Y_i) = \mu_i, \;\; \text{Var}(Y_i) = \mu_i + \mu_i^2/\phi$$

Interpretation:

- $\boldsymbol{\beta}$: log rate ratios (same as Poisson).
- ϕ: dispersion parameter. $\phi \to \infty \Rightarrow$ Poisson.
- Variance is **quadratic** in μ (vs linear for quasi-Poisson).
- Has proper likelihood — can use AIC/LRT.

5. Zero-Inflated Poisson (ZIP, L13)

Model: Latent indicator Z_i for "structural zero":

$$P(Z_i = 0) = \pi_i, \;\; Y_i | Z_i = 0 \equiv 0, \;\; Y_i | Z_i = 1 \sim \text{Poisson}(\lambda_i)$$

$$\log \lambda_i = \mathbf{x}_i^T \boldsymbol{\beta}, \;\; \text{logit}(\pi_i) = \mathbf{z}_i^T \boldsymbol{\gamma}$$

Interpretation:

- $\boldsymbol{\beta}$ (count part): log rate ratios for **non-structural-zero** subjects.
- $\boldsymbol{\gamma}$ (zero part): log-odds ratios for being a structural zero (always-zero) per unit z.
- $P(Y_i = 0) = \pi_i + (1-\pi_i) e^{-\lambda_i}$.

6. Contingency Table: Poisson Model (L14)

Model: Cell counts $Y_{ij} \sim \text{Poisson}(\mu_{ij})$ (no margin fixed).

- **Saturated:** $\hat{\mu}_{ij} = y_{ij}$ (IJ params).
- **Independence (additive):**

$$\log \mu_{ij} = \gamma + \alpha_i + \beta_j, \;\; \sum \alpha_i = \sum \beta_j = 0$$

Interpretation:

- γ: overall log mean.
- α_i: log multiplicative effect of row i.
- β_j: log multiplicative effect of column j.
- No row–column interaction $\Leftrightarrow$ row and column variables are **independent**.

7. Contingency Table: Multinomial (L14)

Model: Total $n = \sum y_{ij}$ fixed.

$$f(\mathbf{y}|n) = n! \prod_{i,j} \frac{\pi_{ij}^{y_{ij}}}{y_{ij}!}, \;\; \sum \pi_{ij} = 1$$

Independence: $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$.

Interpretation:

- π_{ij}: joint probability of cell (i,j).
- $\pi_{i\cdot}, \pi_{\cdot j}$: marginal probabilities. MLEs: $\hat{\pi}_{i\cdot} = Y_{i+}/n$.
- $LR \sim \chi^2_{(I-1)(J-1)}$ — same as Poisson independence test.

8. Contingency Table: Product Multinomial (L14)

Model: Row totals n_{i+} fixed (e.g., case-control). Each row independently multinomial.

- **Saturated:** $\hat{\pi}_{j|i} = y_{ij}/n_{i+}$.
- **Homogeneity:** $\pi_{j|i} = \pi_{\cdot j}$ for all i.

Interpretation:

- $\pi_{j|i}$: conditional probability of column j given row i.
- Homogeneity $\Leftrightarrow$ row classification doesn't affect column distribution (independence).
- $df = (I-1)(J-1)$ as before.

9. Marginal Linear Model for Longitudinal (L15-16)

Model: Repeated measurements Y_{ij} on subject i, time j.

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \boldsymbol{\epsilon}_i, \;\; \boldsymbol{\epsilon}_i \sim N(\mathbf{0}, \boldsymbol{\Sigma})$$

$$E(Y_{ij}) = \mathbf{X}_{ij}^T\boldsymbol{\beta}, \;\; \mathrm{Var}(\mathbf{Y}_i) = \boldsymbol{\Sigma}$$

Interpretation:

- $\boldsymbol{\beta}$: **population-averaged** effect — change in mean response per unit x across the population.
- $\boldsymbol{\Sigma}$: within-subject covariance structure (CS, AR1, Toeplitz, exp, unstructured).

Covariance Structures (different $\boldsymbol{\Sigma}$):

- **Compound Sym:** σ^2 on diag, $\sigma^2\rho$ off-diag (2 params).
- **Toeplitz:** $\mathrm{corr}(Y_{ij}, Y_{i,j+k}) = \rho_k$ (n params, equal spacing).
- **AR(1):** $\rho_k = \rho^k$ (2 params, equal spacing).
- **Exponential:** $\mathrm{corr} = \rho^{|t_j - t_k|}$ (uneven spacing).
- **Unstructured:** all entries free.

10. Linear Mixed Effects: Random Intercept (L17)

Model:

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}^T\boldsymbol{\beta} + b_i + \epsilon_{ij}$$

$$b_i \sim N(0, \sigma_b^2), \;\; \epsilon_{ij} \sim N(0, \sigma^2), \;\; \text{indep}$$

Interpretation:

- $\boldsymbol{\beta}$: **fixed effect** — population-average change in Y per unit x (also = subject-specific, since random part is just an intercept shift).
- b_i: **subject-specific deviation** from population intercept.
- σ_b^2: between-subject variance; σ^2: within-subject (residual).
- **ICC** $= \sigma_b^2/(\sigma_b^2 + \sigma^2)$: proportion of total variance from between-subject heterogeneity.
- Induces **compound symmetry** marginal covariance: $\mathrm{Cov}(Y_{ij}, Y_{ik}) = \sigma_b^2$.

11. Linear Mixed Effects: Random Intercept and Slope (L17)

Model:

$$Y_{ij} = \beta_1 + \beta_2 t_{ij} + b_{1i} + b_{2i}t_{ij} + \epsilon_{ij}$$

$$\mathbf{b}_i = (b_{1i}, b_{2i})^T \sim N(\mathbf{0}, \mathbf{G}), \;\; \mathbf{G} = \begin{pmatrix} g_{11} & g_{12} \\ g_{12} & g_{22} \end{pmatrix}$$

Interpretation:

- β_1: population mean intercept.
- β_2: population mean slope (rate of change over time).
- b_{1i}: subject i's deviation in baseline level.
- b_{2i}: subject i's deviation in rate of change over time.
- g_{11}, g_{22}: between-subject variance in intercept/slope.
- g_{12}: covariance between intercept & slope (e.g., negative g_{12} = subjects starting higher tend to decline faster).
- $\mathrm{Var}(Y_{ij}) = g_{11} + 2t_{ij}g_{12} + t_{ij}^2 g_{22} + \sigma^2$ (depends on time).

12. General Linear Mixed Model (L17)

Model: $\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \boldsymbol{\epsilon}_i$

$$\mathbf{b}_i \sim N(\mathbf{0}, \mathbf{G}), \;\; \boldsymbol{\epsilon}_i \sim N(\mathbf{0}, \sigma^2\mathbf{I})$$

$$E(\mathbf{Y}_i|\mathbf{b}_i) = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i, \;\; E(\mathbf{Y}_i) = \mathbf{X}_i\boldsymbol{\beta}$$
$$\mathrm{Var}(\mathbf{Y}_i) = \mathbf{Z}_i\mathbf{G}\mathbf{Z}_i^T + \sigma^2\mathbf{I}$$

Interpretation:

- $\boldsymbol{\beta}$: fixed (population) effects — same for everyone.
- $\mathbf{b}_i$: random subject-specific effects, predicted by **BLUP**.
- $\mathbf{Z}_i$: design matrix selecting which coefficients vary by subject (subset of columns of $\mathbf{X}_i$).

13. GEE: Marginal Model for non-Gaussian Longitudinal (L19)

Model: No full distribution specified.

$$g(\mu_{ij}) = \mathbf{X}_{ij}^T\boldsymbol{\beta}, \;\; \mathrm{Var}(Y_{ij}) = \phi V(\mu_{ij})$$

$$\mathrm{corr}(Y_{ij}, Y_{ik}) = \alpha_{jk} \text{ (working corr)}$$

Estimating equation:

$$\sum_i \mathbf{D}_i^T\boldsymbol{\Sigma}_i^{-1}(\mathbf{Y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$$

Interpretation:

- $\boldsymbol{\beta}$: **population-averaged** — comparing mean response across **subgroups of population** differing by X value.
- E.g., logistic GEE: β_j = log OR comparing two subpopulations differing in X_j by 1 unit.
- ϕ: scale parameter (dispersion).
- α_{jk}: working correlation params (nuisance — robust to misspecification).

Common GEE Specifications:

- **Continuous:** identity link, $\mathrm{Var} = \sigma^2$, AR(1) corr.
- **Count:** log link, $\mathrm{Var} = \phi\mu$, exchangeable corr.
- **Binary:** logit link, $\mathrm{Var} = \mu(1-\mu)$, unstructured corr.

14. Generalized Linear Mixed Model (GLMM, L20)

Model: Conditional on $\mathbf{b}_i$:

$$Y_{ij}|\mathbf{b}_i \sim \text{EF}, \;\; g(E(Y_{ij}|\mathbf{b}_i)) = \mathbf{X}_{ij}^T\boldsymbol{\beta} + \mathbf{Z}_{ij}^T\mathbf{b}_i$$
$$\mathbf{b}_i \sim N(\mathbf{0}, \mathbf{G})$$

Interpretation:

- $\boldsymbol{\beta}$: **subject-specific** effect — change in $g(\mu)$ per unit x for the **same individual** ($\mathbf{b}_i$ held fixed).
- $\mathbf{b}_i$: subject-specific random effect; captures unobserved heterogeneity.
- For nonlinear g (logit/log), $\boldsymbol{\beta}_{GLMM} \neq \boldsymbol{\beta}_{Marginal}$. Typically $|\beta_{GLMM}| > |\beta_{Marginal}|$.
- For linear g (identity/Gaussian), GLMM = LMM and the two coincide.

14a. Logistic GLMM (Random Intercept)

$$\text{logit}(P(Y_{ij} = 1|b_i)) = b_i + \beta_1 + X_{ij}\beta_2, \;\; b_i \sim N(0, \sigma_b^2)$$

- β_2: log OR per unit X for **same subject**.
- For two subjects differing in X by 1, log OR diff $= \beta_2 + b_i - b_{i'}$.

14b. Poisson GLMM (Random Int + Slope)

$$\log E(Y_{ij}|\mathbf{b}_i) = (b_{1i} + \beta_1) + (b_{2i} + \beta_2)t_{ij}$$

- β_2: log rate ratio per unit time, **subject-specific**.
- $\mathbf{b}_i \sim N(\mathbf{0}, \mathbf{G})$: heterogeneity in baseline rate AND time trend.

15. Parametric Survival Distributions (L21)

15a. Exponential

$$f(t) = \lambda e^{-\lambda t}, \;\; S(t) = e^{-\lambda t}, \;\; h(t) = \lambda$$

- λ: **constant hazard rate** (events per unit time).
- Memoryless property; $E(T) = 1/\lambda$.

15b. Weibull

$$S(t) = e^{-\lambda t^\alpha}, \;\; h(t) = \lambda\alpha t^{\alpha-1}, \;\; f(t) = \lambda\alpha t^{\alpha-1}e^{-\lambda t^\alpha}$$

- λ: scale parameter.
- α: shape parameter.
  - $\alpha > 1$: hazard increases with time (e.g., aging).
  - $\alpha < 1$: hazard decreases with time (e.g., infant mortality).
  - $\alpha = 1$: constant hazard $\Rightarrow$ exponential.

16. Kaplan-Meier (Nonparametric, L21)

Model: No parametric form. Estimate $S(t)$ directly.

$$\hat{S}(t) = \prod_{i:t_i \leq t}\left(1 - \frac{d_i}{n_i}\right)$$

Interpretation:

- $\hat{\lambda}_i = d_i/n_i$: estimated conditional probability of dying at t_i given survival to t_i^-.
- $\hat{S}(t)$: probability of surviving past t.
- Step function — drops only at uncensored event times.

17. Accelerated Failure Time (AFT) Model (L23)

Model:

$$\log T_i = \mathbf{X}_i^T\boldsymbol{\beta} + \epsilon_i$$
$$T_i = \exp(\mathbf{X}_i^T\boldsymbol{\beta}) \cdot T_{0i}$$

where $\epsilon_i \sim$ parametric distribution (Weibull, Exponential, Log-normal, etc.).

Interpretation:

- $\boldsymbol{\beta}$: effects directly on the **survival time** scale.
- $\exp(\beta_j)$: **time accelerator/decelerator**. Per unit $\uparrow x_j$, expected event time becomes $\exp(\beta_j)$ times the original.
  - $\beta_j > 0$: time **lengthened** (longer survival).
  - $\beta_j < 0$: time **shortened** (faster failure).
- T_{0i}: baseline lifetime distribution.

18. Cox Proportional Hazards (PH) Model (L23)

Model (semi-parametric):

$$h_i(t) = h_0(t)\exp(\mathbf{X}_i^T\boldsymbol{\beta})$$

Interpretation:

- $h_0(t)$: **baseline hazard** function (nonparametric, unspecified).
- $\boldsymbol{\beta}$: log hazard ratio.
- $\exp(\beta_j)$: **hazard ratio (HR)** for unit $\uparrow x_j$, holding others fixed.
  - $\beta_j > 0$: higher hazard $\Rightarrow$ **shorter** survival.
  - $\beta_j < 0$: lower hazard $\Rightarrow$ longer survival.
- **No intercept** (absorbed into h_0).
- HR is **constant over time** (key PH assumption).
- $\frac{h_i(t)}{h_k(t)} = \exp((\mathbf{X}_i - \mathbf{X}_k)^T\boldsymbol{\beta})$ — does not depend on t.

Estimation: Partial likelihood

$$L(\boldsymbol{\beta}) = \prod_{i'=1}^{k}\frac{\exp(\mathbf{x}_{(i')}^T\boldsymbol{\beta})}{\sum_{j\in R_{(i')}}\exp(\mathbf{x}_j^T\boldsymbol{\beta})}$$

19. Cox PH with Time-Dependent Covariates (L23)

Model:

$$h_i(t) = h_0(t)\exp(\mathbf{x}_i(t)^T\boldsymbol{\beta})$$

Interpretation:

- $\mathbf{x}_i(t)$: covariates that change over time (e.g., age, biomarker, treatment dose).
- β_j: log HR per unit change in x_j **at any given time point**.
- Estimation via partial likelihood with $\mathbf{x}_j(t_{(i')})$ used in risk set at each event time $t_{(i')}$.

20. Stratified Cox PH Model (L23)

Model: When PH fails for stratifier S but holds within each stratum:

$$h_{is}(t) = h_{0s}(t)\exp(\mathbf{x}_i^T\boldsymbol{\beta})$$

Interpretation:

- $h_{0s}(t)$: **stratum-specific baseline hazard** (different per stratum).
- $\boldsymbol{\beta}$: **same across strata** — common HR for $\mathbf{x}_i$ within each stratum.
- $\exp(\beta_j)$: log HR per unit $\uparrow x_j$ (constant within stratum, but baseline varies).
- **Cannot estimate effect of S itself** (no HR for stratifier).
- PH only required **within** strata.

21. Comparison of $\boldsymbol{\beta}$ Interpretations

(scope of effect; meaning of $\exp(\beta_j)$)

- **Poisson reg:** population; rate ratio.
- **Quasi-Poisson:** population; rate ratio (same $\hat{\beta}$ as Poisson, different SE).
- **NB regression:** population; rate ratio.
- **ZIP (count part):** among non-structural-zeros; rate ratio.
- **ZIP (zero part):** log-odds of being structural zero.
- **Marginal LM (GLS):** population mean; depends on link.
- **LMM (linear):** population = subject-specific (coincide for identity link).
- **GEE: population-averaged**; OR/RR (link-dependent).
- **GLMM: subject-specific**; OR/RR conditional on $\mathbf{b}_i$.
- **AFT:** on **survival time** scale; time accelerator $\exp(\beta_j)$.
- **Cox PH:** on **hazard** scale; hazard ratio $\exp(\beta_j)$.

22. Summary: Choose Your Model

Counts:

- $E \approx Var$ → **Poisson**.
- Exposure varies → **Poisson + offset**.
- $Var > E$ (overdispersion) → **Quasi-Poisson** or **NB**.
- Excess zeros → **ZIP / ZINB**.

Cross-classified counts: → Poisson / multinomial / product multinomial GLM (independence test).

Longitudinal continuous:

- Population avg only → **GLS marginal** (covariance pattern).
- Subject-specific trajectories → **LMM** (random int / int+slope).

Longitudinal binary/count:

- Population avg, robust SE → **GEE**.
- Subject-specific → **GLMM**.

Survival:

- $S(t)$ with censoring, no covariates → **Kaplan-Meier**.
- Compare 2 groups overall → **log-rank** test.
- Effect on survival **time** (parametric) → **AFT**.
- Effect on **hazard** (semi-parametric) → **Cox PH**.
- Time-varying covariate → Cox with $\mathbf{x}(t)$.
- One covariate violates PH → **Stratified Cox**.